

EE5502 MOS Devices

CA2 Literature Review Project

Sem 1, 2025/26

Zhu Chunxiang

Dept of Electrical and Computer Engineering

Email: elezhucx@nus.edu.sg

Office: E1-08-08

Phone: 6516 8930

Arrangement

- Group base project with 3 students
(same group as the simulation project)
- Report and Presentation
 - ✓ Submit one report per group to CANVAS system
 - ✓ Submit one recorded video presentation to CANVAS system
- Deadline: **Sunday 16 November 2025 23:59**
- Penalty will be applied for late submission
- Email me when having questions.
- Don't copy the sentences, write your own words.

Report Format

- A4 paper, no more than 15 pages, 12 font size, times new roman, double space (excluding references and cover page)
- Each group only needs to submit one softcopy into CANVAS folder (in word or pdf format).
- Clearly state the contribution of each team member in the report.

Presentation Format

- Presentation time should be no more than 15 mins for each group.
- Each member in the group needs to present and to introduce yourself before making presentation.
- Submit one recorded video presentation into CANVAS system before the deadline.
- Clearly introduce your name before each of you making the presentation.
- One example of video recording is to use ZOOM.

How to decide/ choose a topic

- Requirements:
 - ✓ The topic shall be related to MOSFETs.
- Brainstorming within group:
- Some examples:
 - ✓ MOSFETs with new materials/structures: SOI MOSFETs, FinFETs, IGZO TFTs, GaN HEMTs, Si Nanowire FETs, 2D FETs, Negative Capacitance FETs, ...
 - ✓ Various effects in FETs: short channel effects, velocity saturation, hot carrier effects, NBTI, ...
 - ✓ Various memories: DRAM, SRAM, Flash memory, emerging memories based on FETs, ...

How to search research papers:

- Use “Google Scholar” (add lib link: NUS)
- Digital lib, NUS.
- Prefer for papers published in recent years and in high quality journals.
- Example of high quality journals:
 - From well-known journals: Nature, Science and their sub-journals (Nature Communications, etc.), Nano Letters, ACS Nano, Advanced Materials, Advanced Functional Materials, ...
 - From Professional Society: IEEE, ACS, APS, ...
 - From well established publisher: Springer, Taylor & Francis, Elsevier, Wiley, ...
- You may first read review article to understand the overall picture and then move on to read a few relevant research articles in detail.